Sample mean, σ known -> Z

1 RECOGNISE IT

- Probability concerns $\bar{X}$, not one X.
- Population σ is explicitly known.
- Population normal, or CLT justifies approximate normality.

2 DECISION / METHOD

1. Check the $\bar{X}$ normality condition.
2. Compute SE=σ/√n.
3. Standardise the $\bar{X}$ boundary.
4. Use left/right/between normal area.

3 FORMULA

$\bar{X} \sim N(\mu, \sigma^2/n)$

$z=(\bar{x}-\mu)/(\sigma/\sqrt{n})$

4 FORMULA SHEET VS WHAT I KNOW

Gives: $\bar{X}$, SE=σ/√n, Z when σ known.
Still know: the event direction and probability conversion.

5 EXCEL ROUTE

Direct left tail:
`=NORM.DIST(xbar,mu,sigma/SQRT(n),TRUE)`
Right = 1 minus; between = upper minus lower.

6 TINY WORKED EXAMPLE

μ=50, σ=12, n=36. For P($\bar{X}$<53), SE=2 and z=1.5, so probability is about 0.9332.

7 TOP EXAM TRAPS

- Using σ instead of σ/√n.
- Using t despite known population σ.
- Applying a normal model without population normality or adequate n.

8 SAY ALOUD

Known population σ means the sample mean uses a Z model with standard error σ over square root n.

9 EXIT CHECK

What Excel formula gives P($\bar{X}$>53) for the tiny example?

ONE-GLANCE VISUAL ROUTE

